

Sri Lanka Institute of Information Technology



Software Engineering | SE2030

Sprint Planning and Backlog Management for Web-based Bus Ticket Reservation System

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Identified Personas

1. **Company Owner:** Makes strategic decisions and monitors company performance.
2. **System Administrator:** Manages system operations and resolves technical issues.
3. **Bus Driver:** Drives buses safely, adhering to routes and schedules.
4. **Bus Conductor:** Validates tickets and assists passengers during boarding.
5. **Bus Owner:** Provides buses and monitors their usage.
6. **Passenger:** Books tickets and travels using the bus service.

User stories

Bus Owner

- ✓ As a Bus Owner, I want to add new buses to the system, so that I can expand my fleet.
- ✓ As a Bus Owner, I want to view details of all buses, so that I can monitor the operational fleet.
- ✓ As a Bus Owner, I want to update bus details such as seating capacity and license information, so that records are always accurate.
- ✓ As a Bus Owner, I want to remove buses from the system, so that only active buses are listed.

Bus Conductor

- ✓ As a Bus Conductor, I want to create new on-board tickets upon passenger request, only if seats are available, so that I can accommodate last-minute travelers.
- ✓ As a Bus Conductor, I want to view the passenger list for each journey, including where they board and all relevant details, so that I can manage passengers efficiently.
- ✓ As a Bus Conductor, I want to update passenger status to indicate when they board and when they alight, so that journey records remain accurate.
- ✓ As a Bus Conductor, I want to cancel tickets and remove passengers from the journey if they do not show up, so that their seats can be reassigned.

Bus Driver

- ✓ As a Bus Driver, I want to start a trip log, so that I can record journey details.
- ✓ As a Bus Driver, I want to view my assigned route details, so that I can follow the correct schedule.
- ✓ As a Bus Driver, I want to update trip status to reflect progress, so that operations staff can track the journey.
- ✓ As a Bus Driver, I want to cancel a trip log, so that records are accurate when a trip is not completed.

Passenger

- ✓ As a Passenger, I want to book a ticket for my journey, so that I can secure a seat in advance.
- ✓ As a Passenger, I want to view my booked ticket status and e-ticket, so that I can confirm my travel plans.
- ✓ As a Passenger, I want to update my booking details, so that I can make changes if my plans change.
- ✓ As a Passenger, I want to cancel my booking, so that I can free up the seat if I no longer need it.

System Administrator

- ✓ As a System Administrator, I want to create new schedules for bus drivers and conductors, so that journeys are well organized.
- ✓ As a System Administrator, I want to view all staff schedules, so that I can monitor assignments.
- ✓ As a System Administrator, I want to update driver and conductor schedules, so that changes can be managed efficiently.
- ✓ As a System Administrator, I want to delete outdated schedules, so that the system remains current.

Company Owner

- ✓ As a Company Owner, I want to add new routes, so that I can expand service coverage.
- ✓ As a Company Owner, I want to view all routes in a structured format, so that I can monitor operations.
- ✓ As a Company Owner, I want to update route details, so that the information is always correct.
- ✓ As a Company Owner, I want to delete routes, so that only active routes are listed.

SCRUM Roles

1. Product Owner:
 - Company Owner - Jayawardhane T.D (IT24103015)
2. SCRUM Master:
 - Zikra G.F (IT24104374)
3. Development Team
 - Full Stack - Kalendra G.K.A (IT24102866)
 - Full Stack - Daskon W.W.D.A.M.P.B(IT24103309)
 - Full Stack - Dissanayake Y.I (IT24103931)
 - Full Stack - Bandara C.V (IT24102501)

Product Backlog

ID	User Story	Priority
PBL -1	As a Bus Owner, I want to add new buses to the system, so that I can expand my fleet.	High
PBL-2	As a Bus Owner, I want to view details of all buses, so that I can monitor the operational fleet.	Medium
PBL-3	As a Bus Owner, I want to update bus details such as seating capacity and license information, so that records are always accurate.	Medium
PBL-4	As a Bus Owner, I want to remove buses from the system, so that only active buses are listed.	Low
PBL-5	As a Bus Conductor, I want to create new on-board tickets upon passenger request, only if seats are available, so that I can accommodate last-minute travelers.	High
PBL-6	As a Bus Conductor, I want to update passenger status to indicate when they board and when they alight, so that journey records remain accurate.	High
PBL-7	As a Bus Conductor, I want to view the passenger list for each journey, including where they board and all relevant details, so that I can manage passengers efficiently.	High
PBL-8	As a Bus Conductor, I want to cancel tickets and remove passengers from the journey if they do not show up, so that their seats can be reassigned.	Medium
PBL-9	As a Bus Driver, I want to start a trip log, so that I can record journey details.	High
PBL-10	As a Bus Driver, I want to view my assigned route details, so that I can follow the correct schedule.	Medium
PBL-11	As a Bus Driver, I want to update trip status to reflect progress, so that operations staff can track the journey.	Medium
PBL-12	As a Bus Driver, I want to cancel a trip log, so that records are accurate when a trip is not completed.	Low
PBL-13	As a Passenger, I want to book a ticket for my journey, so that I can secure a seat in advance.	High

PBL-14	As a Passenger, I want to update my booking details, so that I can make changes if my plans change.	Medium
PBL-15	As a Passenger, I want to cancel my booking, so that I can free up the seat if I no longer need it.	Medium
PBL-16	As a Passenger, I want to view my booked ticket status and e-ticket, so that I can confirm my travel plans.	Low
PBL-17	As a System Administrator, I want to create new schedules for bus drivers and conductors, so that journeys are well organized.	High
PBL-18	As a System Administrator, I want to view all staff schedules, so that I can monitor assignments.	Medium
PBL-19	As a System Administrator, I want to update driver and conductor schedules, so that changes can be managed efficiently.	Medium
PBL-20	As a System Administrator, I want to delete outdated schedules, so that the system remains current.	Low
PBL-21	As a Company Owner, I want to add new routes, so that I can expand service coverage.	High
PBL-22	As a Company Owner, I want to view all routes in a structured format, so that I can monitor operations.	Medium
PBL-23	As a Company Owner, I want to update route details, so that the information is always correct.	Medium
PBL-24	As a Company Owner, I want to delete routes, so that only active routes are listed.	Low

Sprint Planning

1st Sprint – week 3-7

Sprint ID	User Story	Priority
SP1-1	As a Company Owner, I want to add new routes, so that I can expand service coverage.	High
SP1-2	As a Passenger, I want to book a ticket for my journey, so that I can secure a seat in advance.	High
SP1-3	As a Bus Conductor, I want to view the passenger list for each journey, including where they board and all relevant details.	High
SP1-4	As a Bus Owner, I want to add new buses to the system, so that I can expand my fleet.	High
SP1-5	As a Bus Driver, I want to start a trip log, so that I can record journey details.	High
SP1-6	As a System Administrator, I want to create new schedules for bus drivers and conductors, so that journeys are well organized.	High

Sprint Backlog

Task 1: Add New Routes (High)

Task ID	Task	Est. Hours
T-01.1	Design UI form for adding new routes	5
T-01.2	Create backend API to store route details	6
T-01.3	Validate route details (distance, stops, timings)	3
T-01.4	Save route in database with status “Active”	2
T-01.5	Test add route functionality	3

Task 2: Book a Ticket (High)

Task ID	Task	Est. Hours
T-02.1	Design booking UI (route, date, seat selection)	6
T-02.2	Implement seat availability check in backend	5
T-02.3	Integrate payment gateway (Stripe/PayPal/local)	8
T-02.4	Process payment and generate booking reference	5
T-02.5	Auto-generate e-ticket PDF after successful payment	4
T-02.6	Send booking confirmation via email/SMS	3
T-02.7	Test booking + payment flow	4

Task 3: View Passenger List for Each Journey

Task ID	Task	Est. Hours
T-03.1	Create a UI that can view passenger list for each journey displaying passenger name, seat number, boarding point, destination, and ticket status.	4
T-03.2	Implement backend logic to fetch passenger details from the database based on journey ID.	5
T-03.3	Add sorting and filtering options (by seat number, boarding point, or ticket status) for easy navigation.	5
T-03.4	Test with multiple journeys and passenger records to ensure data accuracy.	5

Task 4: Add New Buses

Task ID	Task	Est. Hours
T-04.1	Design UI form for adding new bus details	5
T-04.2	Implement backend service to save bus details	6
T-04.3	Validate bus registration number and capacity	3
T-04.4	Store bus information in database	2
T-04.5	Test add bus functionality	3

Task 5: Start Trip Log

Task ID	Task	Est. Hours
T-05.1	Design trip log start UI	4
T-05.2	Backend to initiate trip log	5
T-05.3	Capture driver, bus, route, time details	3
T-05.4	Store log entry in database	2
T-05.5	Test trip log start process	3

Task 6: Create Schedules

Task ID	Task	Est. Hours
T-06.1	Design UI for schedule creation	5
T-06.2	Backend API for creating driver/conductor schedules	6
T-06.3	Add validations (conflicts, shift limits)	3
T-06.4	Store schedule in database	2
T-06.5	Test schedule creation	3

2nd Sprint – Week 7 -10

Sprint ID	User Story	Priority
SP2-1	PBL-5 – As a Bus Conductor, I want to create new on-board tickets upon passenger request, only if seats are available.	High
SP2-2	PBL-6-As a Bus Conductor, I want to update passenger status to indicate when they board and when they alight, so that journey records remain accurate	High
SP2-3	PBL-2 – As a Company Owner, I want to update and delete route details, so that the information is always correct.	Medium
SP2-4	PBL-6 – As a Passenger, I want to cancel my booking, so that I can free up the seat if I no longer need it.	Medium
SP2-5	PBL-8 – As a Bus Conductor, I want to cancel tickets and remove passengers from the journey if they do not show up, so that their seats can be reassigned.	Medium

Task 1: Create New On-Board Tickets (High)

Task ID	Task	Est. Hours
T-1.1	Design UI for creating on-board tickets	5
T-1.2	Implement backend API to create tickets	6
T-1.3	Check seat availability before issuing ticket	4
T-1.4	Assign seat number automatically	3
T-1.5	Generate ticket receipt/e-ticket	3
T-1.6	Test end-to-end on-board ticket creation	3

Task 2: Update Passenger Status (High)

Task ID	Task	Est. Hours
T-2.1	update a passenger list view for the journey with “Boarded” and “Alighted” buttons	4
T-2.2	Update passenger status in the database when actions are taken.	5
T-2.3	Ensure real-time sync so the system reflects updated statuses instantly.	3
T-2.4	Test with multiple passengers and journeys.	2

Task 3: Update/Delete Route Details (Medium)

Task ID	Task	Est. Hours
T-02.1	Design UI for updating/deleting route details	5
T-02.2	Backend API for updating route information	6
T-02.3	Backend API for deleting routes safely	4
T-02.4	Validate updates (conflicts, dependencies)	3
T-02.5	Test update/delete route functionality	3

Task 4: Cancel Booking (Medium)

Task ID	Task	Est. Hours
T-03.1	Design UI for passenger to cancel booking	4
T-03.2	Implement backend API to process cancellation	5
T-03.3	Free up seat for other passengers	3
T-03.4	Update booking status in database	2
T-03.5	Test cancellation functionality	3

Task 5: Cancel Tickets if Passenger Does Not Show Up (Medium)

Task ID	Task	Est. Hours
T-04.1	Design UI option to cancel ticket	4
T-04.2	Backend API to remove passenger from journey	5
T-04.3	Update seat availability for reassignment	3
T-04.4	Notify passenger (if needed)	2
T-04.5	Test ticket cancellation process	3

3rd Sprint – Week 11-12

Sprint ID	User Story	Priority
SP3-1	PBL-3 – As a Company Owner, I want to view all routes in a structured format, so that I can monitor operations.	Medium
SP3-2	PBL-12 – As a Bus Driver, I want to view my assigned route details, so that I can follow the correct schedule.	Medium
SP3-3	PBL-17 – As a Bus Driver, I want to update trip status to reflect progress.	Medium
SP3-4	PBL-18 – As a Bus Driver, I want to cancel a trip log, so that records are accurate when a trip is not completed.	Medium
SP3-5	PBL-19 – As a System Administrator, I want to view all staff schedules, so that I can monitor assignments.	Medium

Task 1: View All Routes (Medium)

Task ID	Task	Est. Hours
T-01.1	Design UI to display all routes in table/grid	4
T-01.2	Backend API to fetch route data	5
T-01.3	Add filters (by city, distance, bus type)	3
T-01.4	Ensure real-time data update	2
T-01.5	Test route viewing functionality	3

Task 2: View Assigned Route Details (Medium)

Task ID	Task	Est. Hours
T-2.1	Design driver interface to view route	4
T-2.2	Fetch assigned route details from backend	5
T-2.3	Display stops, timings, and passenger info	3
T-2.4	Add refresh feature for updates	2
T-2.5	Test assigned route view functionality	3

Task 3: Update Trip Status (Medium)

Task ID	Task	Est. Hours
T-3.1	Design UI to update trip progress	4
T-3.2	Implement backend API to save trip status	5
T-3.3	Track current stop, delay, or completion	3
T-3.4	Sync updates to admin dashboard	2
T-3.5	Test trip status update flow	3

Task 4: Cancel Trip Log (Medium)

Task ID	Task	Est. Hours
T-4.1	Design UI option to cancel trip log	4
T-4.2	Backend API to remove or mark trip as cancelled	5
T-4.3	Notify admin and drivers	3
T-4.4	Update related passenger/trip records	3
T-4.5	Test trip log cancellation	2

Task 5: View Staff Schedules (Medium)

Task ID	Task	Est. Hours
T-5.1	Design UI to view driver/conductor schedules	4
T-5.2	Backend API to fetch schedules	5
T-5.3	Add filters by date/shift/staff	3
T-5.4	Sync schedules with updates from admin	2
T-5.5	Test schedule viewing functionality	3

4th Sprint – 13th week

Sprint ID	User Story	Priority
SP4-1	PBL-5 – As a Passenger, I want to view my booked ticket status and e-ticket.	Low
SP4-2	PBL-10 – As a Bus Owner, I want to update bus details such as seating capacity and license information.	Low
SP4-3	PBL-14 – As a System Administrator, I want to update driver and conductor schedules.	Low
SP4-4	PBL-20 – As a Passenger, I want to update my booking details.	Low
SP4-5	PBL-21 – As a System Administrator, I want to delete outdated schedules.	Low

Task 1: View Booked Ticket Status & E-Ticket (Low)

Task ID	Task	Est. Hours
T-01.1	Design UI for viewing booked ticket and e-ticket	4
T-01.2	Implement backend API to fetch ticket information	5
T-01.3	Display seat number, route, date/time, and QR code	3
T-01.4	Add print and download options for e-ticket	2
T-01.5	Test ticket status and e-ticket display	3

Task 2: Update Bus Details (Low)

Task ID	Task	Est. Hours
T-2.1	Design UI for updating bus details (capacity, license)	4
T-2.1	Implement backend API to update bus information	5
T-2.3	Validate input fields (capacity limits, license format)	3
T-2.4	Save updated bus information in database	2
T-2.5	Test bus detail update functionality	3

Task 3: Update Driver/Conductor Schedules (Low)

Task ID	Task	Est. Hours
T-3.1	Design UI to edit schedules for drivers/conductors	4
T-3.2	Backend API to update schedule data	5
T-3.3	Validate schedule changes to prevent conflicts	3
T-3.4	Save updated schedules in the database	2
T-3.5	Test schedule update feature	3

Task 4: Update Booking Details (Low)

Task ID	Task	Est. Hours
T-4.1	Design UI to allow passengers to edit bookings	4
T-4.2	Backend API to update booking details	5
T-4.3	Validate seat availability if route/date changes	3
T-4.4	Save updated booking in database	2
T-4.5	Send updated e-ticket or notification to passenger	3
T-4.6	Test booking update functionality end-to-end	3

Task 5: Delete Outdated Schedules (Low)

Task ID	Task	Est. Hours
T-5.1	Design admin UI for viewing and selecting outdated schedules	4
T-5.2	Implement backend API to delete schedules	5
T-5.3	Validate that deletion does not affect ongoing trips	3
T-5.4	Update database to mark schedules as deleted or removed	2
T-5.5	Test deletion process and check audit logs	3